28 Volt Power System (Mod S.O.O. 6283) – General data

1. General

There is a 28 volt power outlet configuration available in the cabin to provide power to peripherals: cell phone, laptops, etc.

2. Description

There is a single 28 volt power outlet that may be installed on the forward right hand side of the STA 332 bulkhead in the cabin. The outlet has a maximum load of 5 amps.

28 volt DC Power is provided from the right bus, through a 5 amp circuit breaker (CB18) located on the power distribution circuit box.

3. Removal/Installation

A. Removal of 28V Power Outlet

1) Ensure the aircraft electrical system is OFF.

2) Open the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

3) Remove the countersunk screws from the outlet mounting plate.

4) Remove the outlet assembly from the interior paneling.

5) Gently remove electrical connectors from the rear terminals of the outlet. Note which connector came off the center terminal and which came off the outer terminal.

6) Unthread the two metallic locknuts from the back of the outlet.

7) Remove outlet by pulling it out of the front of the mounting hole.

Note Stow the connectors as required.

B. Installation of 28V Power Outlet

1) Ensure the aircraft electrical system is OFF.

2) Ensure the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

3) Install the outlet in mounting panel from the front of the panel.

4) Thread the two metallic locknuts onto the back of the outlet and tighten. Ensure the outlet is secured in the panel.

5) Install the electrical connectors onto the appropriate terminals on the rear of the outlet.

6) Install outlet assembly from the interior paneling.

7) Install the countersunk screws from the outlet mounting plate.

8) Remove the safety clips and close the CB18 circuit breakers located on the power distribution circuit box and fit safety clips. (Refer to 24-60-11, Power Distribution and Generator Control Box (Mod 6/2028) – General Data).

9) Using a voltmeter, measure the voltage between the center pin and the outer ring of the outlet. Ensure the 'com' lead is connected to the outer ring. Confirm the voltage is +28V (± 1.0V).

4. Troubleshooting

A. Ensure CB18 circuit breakers located on the power distribution circuit box are closed.

B. Visually inspect the outlet connectors for damaged wiring or contacts. Ensure electrical connectors on outlet are fully seated and secure.

C. If troubleshooting at a wire level is required, refer to the applicable wiring diagrams in PSM 1-64-2W.